

A REFERENCE FOR DEVELOPERS

Technical Terms Cookbook

A plain-English glossary of the terms across this whole library — backend, AI/ML, LLMs & agents, data, MLOps & cloud, and freelancing — each explained in a sentence or two.

Glossary

Backend

AI / ML

LLMs & Agents

Data

MLOps & Cloud

Freelancing

Plain English

Written as a fast lookup: skim a domain, or Ctrl-F a term you hit in another cookbook and get a clear, jargon-free explanation.

Each entry answers 'what is it, in one breath' — enough to understand and keep reading, with a pointer to the book that goes deep.

Edition 2026 · define it, don't drown in it

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How to Use This Book

Every other cookbook in this library throws terms at you — *idempotent*, *gradient descent*, *RAG*, *Job Success Score*. This one is the dictionary. It collects the vocabulary from across the whole library and explains each term in **plain English, in a sentence or two** — enough to understand it and keep going, without a rabbit hole.

It's built for two uses:

1. **Skim a domain** before or during a track to get the lay of the land.
2. **Look up a term** (Ctrl-F / search) the moment you hit one you don't know in another book, get a quick answer, and get back to reading.

Definitions favour intuition over precision — they're the "what is this, in one breath" version, with the deep treatment living in the relevant cookbook. It's a map, not the territory.

How it's organized

Terms are grouped by domain, roughly in the order the library builds them:

- **General programming & CS** — the fundamentals.
- **Backend & web** — APIs, databases, infra basics.
- **Data** — analysis and engineering.
- **Machine learning** — classic ML.
- **Deep learning** — neural networks.
- **LLMs** — language models and their vocabulary.
- **Agents & automation** — tools, MCP, n8n.
- **MLOps & cloud** — shipping and scaling.
- **Freelancing** — the business terms.
- **Acronyms** — a quick-expand cheat table at the end.

HOW TO READ IT

Don't read it cover to cover — it's a reference. Use it *alongside* the other cookbooks: when a term stops you, jump here, get the one-line version, and return. Each domain points at the book that goes deep, so a quick definition can become a full understanding whenever you want it.

WHY A GLOSSARY MATTERS

Half of feeling lost in a new field is just unfamiliar words for familiar ideas. A gradient is a slope; a token is a chunk of text; a container is a lightweight box for an app. Once the vocabulary stops being a wall, the concepts underneath are usually simpler than the jargon made them sound. That's the whole job of this book: turn the wall into a doorway.

Part 1 · General Programming & Computer Science

The foundational vocabulary everything else builds on.

1.1 Core concepts

Term	What it means
Algorithm	A step-by-step procedure to solve a problem.
Data structure	A way of organizing data so it's efficient to use (array, list, map, tree...).
Function	A named, reusable block of code that takes inputs and returns an output.
Variable	A named box that holds a value.
Class / Object	A blueprint (class) and an instance made from it (object) — data plus the operations on it.
OOP	Object-Oriented Programming — organizing code around objects.
Recursion	A function that calls itself to solve a smaller version of the problem.
Big-O	A shorthand for how an algorithm's cost grows with input size (e.g. $O(n)$ = linear).
Syntax	The grammar rules of a programming language.
Compiler / Interpreter	Turns your code into something the machine runs — ahead of time (compiler) or line-by-line (interpreter).
Runtime	The environment your code runs in (e.g. Node.js, the Python interpreter).
Framework	A structured toolkit you build <i>inside</i> (it calls your code).
Library	Reusable code you call <i>from</i> your program.

1.2 Data types

Term	What it means
Integer / Float	A whole number / a number with a decimal point.
String	Text.
Boolean	True or false.
Array / List	An ordered collection of values.
Hash map / Dictionary	Key→value lookups (a phone book).
Stack / Queue	Last-in-first-out / first-in-first-out collections.
Tree / Graph	Hierarchical / network-shaped data structures.

Term	What it means
Null / None	The absence of a value.

1.3 Working with code

Term	What it means
IDE	Integrated Development Environment — a code editor with built-in tools.
CLI / Terminal	Command-Line Interface — driving the computer by typing commands.
Git	The standard version-control system for tracking code changes.
Repository (repo)	A project's tracked codebase.
Commit	A saved snapshot of changes.
Branch / Merge	A parallel line of work / combining it back in.
Dependency	External code your project relies on.
Package manager	Installs and manages dependencies (npm, pip, uv).
Open source	Code with a public license anyone can use/read.
Bug / Debugging	A defect / the process of finding and fixing it.
Refactor	Improving code's structure without changing what it does.
Environment variable	A config value passed to a program from outside (not hard-coded).

1.4 Runtime behaviour

Term	What it means
Concurrency	Handling many tasks in overlapping time periods.
Thread / Process	Units of execution — lightweight (thread) or isolated (process).
Async / Asynchronous	Not waiting: start work, do other things, handle the result later.
Latency	How long one operation takes (delay).
Throughput	How much work gets done per unit time.
Cache	A fast store of recently/frequently used data to avoid recomputing.
Idempotent	Safe to run more than once with the same effect as running once.

Part 2 · Backend & Web

The vocabulary of servers, APIs, and the systems behind an app. (*Deep dives: the Python, Laravel, Rust, Redis, Queues, Mailpit cookbooks.*)

2.1 The web basics

Term	What it means
Client / Server	The requester (browser, app) / the machine that answers.
Frontend / Backend	The user-facing part / the behind-the-scenes logic and data.
HTTP / HTTPS	The protocol browsers and servers talk over (S = encrypted).
Request / Response	You ask (request); the server answers (response).
API	Application Programming Interface — a defined way for programs to talk to each other.
REST	A common style for HTTP APIs using URLs and verbs (GET/POST/PUT/DELETE).
Endpoint	A specific URL an API exposes.
CRUD	Create, Read, Update, Delete — the basic data operations.
JSON / YAML	Text formats for structured data (JSON for APIs, YAML for config).
Webhook	An HTTP call <i>a service sends you</i> when an event happens (a reverse API call).

2.2 Data & persistence

Term	What it means
Database	Organized, persistent storage for data.
SQL / NoSQL	Relational (tables, SQL) / non-relational (documents, key-value...) databases.
Query	A request for data from a database.
ORM	Object-Relational Mapper — lets you use database rows as objects in code.
Migration	A versioned change to the database schema.
Index	A structure that makes lookups fast (like a book's index).
Transaction	A group of operations that all succeed or all fail together.
Redis	An in-memory data store used for caching, queues, and fast lookups.

2.3 Architecture & operations

Term	What it means
Authentication / Authorization	Who you are / what you're allowed to do.
Token / JWT	A signed credential proving identity between requests.
Queue / Worker	A backlog of jobs / a process that runs them in the background (BullMQ, Celery).
Message broker	The middleman that holds and routes jobs/events (Redis, RabbitMQ, Kafka).
Rate limiting	Capping how often a client can call you.
Load balancer	Spreads traffic across multiple servers.
Middleware	Code that runs around each request (auth, logging...).
Environment (dev/staging/prod)	Separate copies of your system for building, testing, and live use.
Container	A lightweight, portable box that packages an app with its dependencies (Docker).
Microservices / Monolith	Many small services / one big app — two ways to structure a backend.
Observability	Being able to see what a running system is doing (logs, metrics, traces).

Part 3 · Data (Analysis & Engineering)

The vocabulary of working with data at rest and in motion. (*Deep dives: the Data Analysis and Data Engineering cookbooks.*)

3.1 Analysis

Term	What it means
DataFrame	An in-memory table (rows and columns) — pandas' core object.
pandas / NumPy	Python libraries for tables / numerical arrays.
EDA	Exploratory Data Analysis — looking around a dataset to understand it.
Distribution	How values in a column are spread out (the shape).
Mean / Median	The average / the middle value.
Correlation	Two things moving together (not necessarily cause).
Outlier	A value far from the rest.
Aggregation	Summarizing many rows into one (sum, count, average).
Group-by	Splitting data into groups and aggregating each.
Join / Merge	Combining two datasets on a shared key.
Pivot	Reshaping data between long and wide layouts.
Normalization	Scaling values to a common range.
Confidence interval	A range the true value plausibly falls in.
p-value	The chance a result is just luck; small = probably real.

3.2 Engineering & pipelines

Term	What it means
ETL / ELT	Extract-Transform-Load / Extract-Load-Transform — the two pipeline orders (ELT is modern).
Pipeline	An automated flow that moves and reshapes data from source to consumer.
Data warehouse	A database optimized for analytics (Snowflake, BigQuery).
Data lake	Cheap storage for raw files of any kind (S3 + files).
Lakehouse	A table layer (Delta/Iceberg) giving warehouse features on lake storage.
Parquet	A compressed, columnar file format — the default for analytics data.
Partitioning	Splitting a big table into chunks (usually by date) so queries read less.

Term	What it means
Schema	The shape of your data — columns and their types.
Batch / Streaming	Processing in scheduled chunks / processing each event as it arrives.
Orchestration	Scheduling and sequencing many pipelines (Airflow, Dagster).
DAG	Directed Acyclic Graph — tasks with dependencies, the unit an orchestrator runs.
Backfill	Re-running a pipeline over past periods.
Idempotent load	A write you can re-run without creating duplicates.
dbt	A tool for defining data transformations as tested, version-controlled SQL.
Spark	An engine for processing data too big for one machine, across a cluster.
Kafka	A durable, replayable log for streaming events between systems.
Data quality	Checks that data is correct, complete, and on time.
Fact / Dimension table	Measurements (orders) / the entities they describe (customers) — a star schema.

Part 4 · Machine Learning

The vocabulary of teaching programs from data. (*Deep dives: the Machine Learning and Math for ML cookbooks.*)

4.1 The core idea

Term	What it means
Machine learning	Programs that learn patterns from examples instead of being explicitly coded.
Model	The function a training process produces — maps inputs to predictions.
Training	The process of fitting a model to data.
Feature	An input variable the model uses.
Label / Target	The correct answer you're trying to predict.
Dataset	The examples you train and test on.
Inference / Prediction	Using a trained model on new data.

4.2 Kinds of learning

Term	What it means
Supervised learning	Learning from labelled examples (predict).
Unsupervised learning	Finding structure with no labels (patterns, clusters).
Reinforcement learning	Learning which actions maximize a reward (act).
Regression	Predicting a number (price, temperature).
Classification	Predicting a category (spam / not spam).
Clustering	Grouping similar items without labels.

4.3 Building & evaluating

Term	What it means
Feature engineering	Crafting the input variables that make a model work.
Train/test split	Holding out data to check the model generalizes.
Cross-validation	Rotating which slice is held out, to test more reliably.
Overfitting	Memorizing the training data; great on it, bad on new data.
Underfitting	Too simple to capture the pattern; bad on everything.

Term	What it means
Bias-variance tradeoff	The balance between too-simple (bias) and too-sensitive (variance).
Hyperparameter	A setting you choose before training (e.g. tree depth, learning rate).
Regularization	Techniques that discourage overfitting.
Loss function	A number measuring how wrong the model is (you minimize it).
Gradient descent	The algorithm that lowers the loss by nudging weights downhill.

4.4 Metrics & tools

Term	What it means
Accuracy	Fraction of predictions that were correct.
Precision / Recall	Of predicted positives, how many were right / of actual positives, how many you caught.
F1 score	A single balance of precision and recall.
Confusion matrix	A table of right vs wrong predictions per class.
ROC-AUC	A measure of ranking quality across thresholds.
scikit-learn	The standard Python library for classic ML.
XGBoost / LightGBM	Gradient-boosted trees — the workhorse for tabular prediction.
Data leakage	Test info sneaking into training — makes metrics lie.
Bandit / A-B test	Adaptive / fixed experiments to compare options.

Part 5 · Deep Learning

The vocabulary of neural networks. (*Deep dive: the Deep Learning cookbook.*)

5.1 The building blocks

Term	What it means
Neural network	A stack of simple math layers that learns complex patterns.
Tensor	An n-dimensional array — the data structure everything flows through.
Layer	One transformation step in the network.
Weight / Parameter	A tunable number the network learns.
Activation function	A non-linear step (e.g. ReLU) that lets networks learn complex shapes.
ReLU	A common activation: pass positives, zero out negatives.
Neuron / Unit	A single weighted-sum-plus-activation.
Embedding	A learned vector representing something (a word, a user) so similar things are near.

5.2 Training

Term	What it means
Forward pass	Running data through the network to get a prediction.
Backpropagation	Computing how each weight affected the error, working backward.
Autograd	Automatic differentiation — the library computes gradients for you.
Gradient	The direction/steepness to change a weight to reduce error.
Epoch	One full pass over the training data.
Batch	A group of examples processed together.
Learning rate	The step size when updating weights — the key knob.
Optimizer	The algorithm that applies weight updates (SGD, Adam).
Adam	A popular, forgiving default optimizer.
Dropout	Randomly ignoring units during training to reduce overfitting.
Batch normalization	Normalizing activations between layers to stabilize training.

5.3 Architectures & practice

Term	What it means
CNN	Convolutional Neural Network — great for images.
RNN / LSTM	Networks for sequences (older; transformers now dominate).
Transformer	The architecture behind modern LLMs; uses attention.
Attention	A mechanism letting a model weigh which inputs matter for each output.
Transfer learning	Starting from a pretrained model and adapting it to your task.
Pretrained model / Backbone	A model already trained on huge data you build on.
PyTorch / TensorFlow	The two main deep-learning frameworks (PyTorch dominates).
GPU	A chip that does the massively parallel math deep learning needs.
state_dict	PyTorch's saved bag of a model's weights.

Part 6 · LLMs (Large Language Models)

The vocabulary of language models. (*Deep dives: the Python book's LLM chapter, the RAG and Fine-Tuning cookbooks.*)

6.1 The basics

Term	What it means
LLM	Large Language Model — an AI trained to predict text, used for chat, code, and more.
Token	A chunk of text (roughly a word-piece) — models read and bill per token.
Tokenization	Splitting text into tokens.
Prompt	The input text you give a model.
System prompt	Instructions that set the model's behaviour for a conversation.
Context window	The maximum amount of text (tokens) a model can consider at once.
Temperature	A randomness dial — low = focused, high = creative.
Inference	Running the model to generate output.
Streaming	Getting the response token-by-token as it's generated.
Hallucination	When a model confidently makes something up.
Few-shot	Giving examples in the prompt to steer the output.

6.2 Working with them

Term	What it means
Structured output	Forcing the model to return data matching a schema (e.g. JSON), not free text.
Function / Tool calling	The model asking your code to run a function (see Agents).
Embedding	A vector representing a piece of text's meaning, for search/similarity.
Vector database	A store that finds items by embedding similarity (Pinecone, Qdrant, pgvector).
Cosine similarity	A measure of how alike two embeddings are (the angle between them).
Prompt engineering	Crafting prompts to get better outputs.
Token/cost tracking	Logging tokens used to control spend.
Prompt caching	Reusing computation for repeated prompt prefixes to cut cost/latency.

6.3 RAG & customizing

Term	What it means
RAG	Retrieval-Augmented Generation — fetch relevant facts and put them in the prompt so the model answers from them.
Chunking	Splitting documents into pieces to embed and retrieve.
Retrieval	Finding the top relevant chunks for a query.
Fine-tuning	Changing a model's weights by training it on your examples.
SFT	Supervised Fine-Tuning — training on input→ideal-output pairs.
LoRA / QLoRA	Cheap fine-tuning that trains a tiny add-on adapter (QLoRA on a compressed model).
Adapter	The small trained add-on LoRA produces.
RLHF / DPO	Preference tuning — teaching a model which answers are <i>better</i> (DPO is the simpler modern way).
Quantization	Shrinking a model (e.g. to 4-bit) to run cheaper, with minor quality loss.
Base vs Instruct model	Raw next-token model / one tuned to follow instructions (chat).
Local model	An open model you run yourself (Ollama, Hugging Face) instead of via API.

Part 7 · Agents & Automation

The vocabulary of LLMs that take actions, and workflow tools. (*Deep dives: the AI Agents and n8n cookbooks.*)

7.1 Agents

Term	What it means
Agent	An LLM in a loop with tools, working toward a goal — it acts, not just answers.
Agent loop (ReAct)	Reason → act (call a tool) → observe the result → repeat until done.
Tool	A function you expose to the model for it to call (search, DB query, API).
Tool / Function calling	The mechanism by which the model requests a tool and you run it.
Memory	What the agent remembers — the conversation, summaries, or retrieved facts.
Planning	Breaking a goal into steps before executing (plan-then-execute).
Multi-agent	Several specialized agents coordinated by a supervisor.
Orchestration	Coordinating the steps/agents and their control flow.
Guardrails	Safety limits — validation, approval gates, action limits.
Prompt injection	Malicious instructions hidden in data the agent reads, trying to hijack it.
Eval / Eval set	A test suite of tasks used to measure a (non-deterministic) agent.
LLM-as-judge	Using a model to grade another model's outputs.
Framework	A library that provides the agent loop and plumbing (LangGraph, etc.).

7.2 MCP

Term	What it means
MCP	Model Context Protocol — an open standard for connecting AI apps to tools and data.
MCP server	A process that exposes tools/resources/prompts over MCP; you use existing ones or write your own.
MCP client / host	The AI app (your agent) that connects to MCP servers and uses their tools.
Tools / Resources / Prompts	What an MCP server exposes: callable functions / readable data / reusable prompt templates.
Transport (stdio / HTTP)	How client and server talk — local process (stdio) or remote (HTTP).

7.3 n8n & workflow automation

Term	What it means
n8n	An open-source, self-hosted workflow-automation tool (Node.js based).
Workflow	A pipeline of connected nodes that runs on a trigger.
Node	One step in a workflow (an action, a transform, a condition).
Trigger	What starts a workflow (a webhook, a schedule, an app event).
Item	A unit of data (a JSON object) flowing between nodes; nodes run once per item.
Expression	Inline logic in a node field, written as <code>{{ ... }}</code> .
Code node	A node where you write JavaScript (or sandboxed Python) directly.
Queue mode	n8n's scaled setup where workers run executions (it's BullMQ underneath).
Credential	n8n's encrypted store for a service's secret, kept out of the workflow.

Part 8 · MLOps & Cloud

The vocabulary of shipping, running, and scaling software and models. (*Deep dives: the MLOps, Docker, and Cloud & Kubernetes cookbooks.*)

8.1 MLOps

Term	What it means
MLOps	The practices for deploying and maintaining ML models in production reliably.
Experiment tracking	Recording each training run's params, metrics, and artifacts (MLflow).
Model registry	The source of truth for model versions and their stage (staging/production).
Versioning (data/model)	Tracking exact data and model versions so results are reproducible (DVC).
Reproducibility	Being able to recreate the same model from the same inputs.
Serving	Exposing a model so other systems can get predictions (an API, or batch).
Drift	The world moving away from training data, so accuracy silently drops.
Training/serving skew	Features computed differently in training vs serving — a silent accuracy killer.
Feature store	A shared system serving consistent features to training and serving.
Continuous training (CT)	Retraining automatically on new data or drift, behind a quality gate.
Quality gate	A check that a new model beats the current one before it's promoted.

8.2 Containers & Docker

Term	What it means
Docker	The standard tool for building and running containers.
Container	A lightweight, isolated box packaging an app with its dependencies.
Image	The read-only template a container is started from.
Dockerfile	The recipe for building an image.
Registry	Where images are stored and shared (Docker Hub).
Volume	Persistent storage that outlives a container.
docker-compose	A file to define and run multiple containers together.

8.3 Kubernetes & cloud

Term	What it means
Kubernetes (k8s)	A system that runs and scales containers across a cluster, self-healing.
Pod	The smallest unit — one (or few) containers running together.
Deployment	Declares "keep N replicas of this running" and self-heals.
Service	A stable address that load-balances across pods.
Ingress	A single HTTP entry point routing to services, with TLS.
ConfigMap / Secret	Injected configuration / secrets, kept out of the image.
Probe (liveness/readiness/startup)	Health checks that gate traffic, restart, or grant boot time.
HPA	Horizontal Pod Autoscaler — adds/removes pod replicas on load.
Cluster autoscaler	Adds/removes machines (nodes) so pods can schedule.
Helm	A package manager for Kubernetes apps (templated, versioned).
IaC / Terraform	Infrastructure as Code — declaring cloud infra in version-controlled files.
Serverless	Running code/containers without managing servers (Cloud Run, Lambda).
CI/CD	Continuous Integration / Delivery — automated testing and deployment.

Part 9 · Freelancing

The business vocabulary of freelance work. (*Deep dive: the Freelancing Success cookbook.*)

9.1 Getting work

Term	What it means
Positioning	Who you help and what problem you solve — your one-line pitch.
Niche	A focused specialty you're known for (fewer competitors, higher rates).
Proposal / Cover letter	Your pitch for a posted job on bidding platforms.
Connects	Upwork's paid tokens you spend to submit proposals.
Gig	A packaged, fixed-scope service listing (Fiverr).
Productized service	A repeatable, fixed-price offer sold over and over.
Vetted network	A platform that screens freelancers and matches them to clients (Toptal).
Portfolio / Spec work	Proof of your skill — real projects, or samples you build to demonstrate it.

9.2 Doing the work

Term	What it means
Scope	The agreed definition of what you'll deliver.
Scope creep	The project quietly growing beyond the agreed scope.
Change order	A priced agreement to do extra work outside scope.
Milestone	A chunk of work with its own deliverable and payment.
Escrow	Platform-held funds released as you deliver — protects both sides.
Discovery	The conversation to understand the client's real need before quoting.
Deliverable	A concrete thing you hand over.

9.3 Pricing & growth

Term	What it means
Hourly / Fixed / Value pricing	Charging for time / a defined deliverable / the outcome's worth.
Rate floor	The minimum hourly rate that's sustainable after fees, taxes, and downtime.
Retainer	A recurring monthly arrangement for ongoing work — stable income.

Term	What it means
Platform fee / Commission	The cut a platform takes from your earnings.
Review / Rating	Client feedback that drives your ranking and trust.
JSS	Job Success Score — Upwork's reputation metric.
Repeat client	A past client who rehires — cheap, high-trust revenue.
Red flag	A warning sign of a bad client or job (vague scope, tiny budget, free "tests").
Off-platform	Working with a platform client directly — often against the rules early on.

Part 10 · Acronym Quick-Expand

Every acronym in the library, expanded — for when you just need the words.

10.1 General & backend

Acronym	Expansion
API	Application Programming Interface
REST	REpresentational State Transfer
CRUD	Create, Read, Update, Delete
HTTP(S)	HyperText Transfer Protocol (Secure)
URL	Uniform Resource Locator
JSON	JavaScript Object Notation
YAML	YAML Ain't Markup Language
SQL	Structured Query Language
ORM	Object-Relational Mapper
JWT	JSON Web Token
CLI	Command-Line Interface
IDE	Integrated Development Environment
OOP	Object-Oriented Programming
CI/CD	Continuous Integration / Continuous Delivery (or Deployment)
CPU / GPU	Central / Graphics Processing Unit
OS	Operating System

10.2 Data, ML & AI

Acronym	Expansion
ML / DL	Machine Learning / Deep Learning
AI	Artificial Intelligence
NLP	Natural Language Processing
CV	Computer Vision
RL	Reinforcement Learning
EDA	Exploratory Data Analysis
ETL / ELT	Extract-Transform-Load / Extract-Load-Transform

Acronym	Expansion
DAG	Directed Acyclic Graph
CNN / RNN	Convolutional / Recurrent Neural Network
SGD	Stochastic Gradient Descent
ROC-AUC	Receiver Operating Characteristic – Area Under Curve
MDP	Markov Decision Process
PPO / DQN / SAC	RL algorithms (Proximal Policy Optimization / Deep Q-Network / Soft Actor-Critic)

10.3 LLMs & agents

Acronym	Expansion
LLM	Large Language Model
RAG	Retrieval-Augmented Generation
MCP	Model Context Protocol
SFT	Supervised Fine-Tuning
RLHF	Reinforcement Learning from Human Feedback
DPO	Direct Preference Optimization
LoRA / QLoRA	Low-Rank Adaptation / Quantized LoRA
PEFT	Parameter-Efficient Fine-Tuning
ReAct	Reason + Act (the agent loop pattern)

10.4 MLOps, cloud & business

Acronym	Expansion
MLOps	Machine Learning Operations
k8s	Kubernetes (k + 8 letters + s)
HPA	Horizontal Pod Autoscaler
IaC	Infrastructure as Code
CT	Continuous Training
DVC	Data Version Control
TLS	Transport Layer Security
MVP	Minimum Viable Product
SLA	Service-Level Agreement
TOS	Terms of Service
JSS	Job Success Score (Upwork)

Acronym	Expansion
DTC	Direct-To-Consumer

YOU'RE SET

Keep this book open in a tab while you work through the others — a term you can name is a term you can learn. When something here sparks "wait, how does that actually work?", that's your cue to open the cookbook it points to and go deep.